

CMPE 49T HW 1 PART 1

Q1 first convolution

$$X = \begin{bmatrix} 1 & -1 & 0 & 2 & -2 \\ 2 & 0 & -1 & 1 & 1 \\ -2 & 1 & 2 & -1 & 3 \\ 1 & -2 & 1 & 2 & -1 \\ 3 & 0 & -1 & 1 & 1 \end{bmatrix}, F_1 = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

We use some padding ($p=1$) and stride $s=1$.

The operation is standard CNN cross-correlation. The output is 5×5 .

Let Y denote the output feature map.

a) Top-left position $Y_{1,1}$

$$\text{win}_{1,1} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 2 & 0 \end{bmatrix}$$

Elementwise multiply and sum with F_1 :

$$Y_{1,1} = (1 \cdot 0) + (0 \cdot 0) + (-2 \cdot 0) + (0 \cdot 0) + (1 \cdot 1) + (0 \cdot (-1)) + (-2 \cdot 0) + (0 \cdot 2) + (1 \cdot 0) = 1$$

b) Next position in the first row $Y_{1,2}$

$$\text{win}_{1,2} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & -1 & 0 \\ 2 & 0 & -1 \end{bmatrix} \Rightarrow Y_{1,2} = (1 \cdot 0) + (0 \cdot 0) + (-2 \cdot 0) + (0 \cdot 1) + (1 \cdot (-1)) + (0 \cdot 0) + (-2 \cdot 2) + (0 \cdot 0) + (1 \cdot (-1)) = -6$$

Others has the same calculation as well, so the final answer is

$$Y = \begin{bmatrix} 1 & -6 & 1 & 5 & -4 \\ 5 & 7 & -9 & 4 & 5 \\ -4 & 4 & 6 & -7 & 0 \\ -1 & -15 & 5 & 1 & -4 \\ 7 & -1 & -7 & 4 & 3 \end{bmatrix}$$

Output dimensions:

$p=1$, $s=1$, kernel 3×3 on 5×5 input, the output is 5×5

Q2 | Leaky ReLU (with $\alpha = 0.1$)

$$\text{LReLU}(x) = \begin{cases} x, & x \geq 0 \\ \alpha x, & x < 0 \end{cases} \quad \text{with } \alpha = 0.1$$

Examples: $\text{LReLU}(5) = 5$ and $\text{LReLU}(-6) = 0.1(-6) = -0.6$

$$A = \text{LReLU}(Y) = \begin{bmatrix} 1 & -0.6 & 1 & 5 & -0.4 \\ 5 & 7 & -0.9 & 4 & 5 \\ -0.4 & 4 & 6 & -0.7 & 0 \\ -0.1 & -1.5 & 5 & 1 & -0.4 \\ 7 & -0.1 & -0.7 & 4 & 3 \end{bmatrix}$$

Q3 | Average Pooling (2×2 window, stride = 1)
 I perform 2×2 average pooling with stride 1 on A. The output size:
 $(5-2)/1 + 1 = 4 \Rightarrow 4 \times 4$

Example explicit pooled value, top-left pooled value is the mean of the 2×2 block.

$$\frac{1 + (-0.6) + 5 + 7}{4} = \frac{12.4}{4} = 3.1 = 3.100$$

Full pooled matrix

$$P = \text{AvgPool}_{2 \times 2, s=1}(A) = \begin{bmatrix} 3.100 & 1.625 & 2.275 & 3.400 \\ 3.900 & 4.025 & 2.100 & 2.075 \\ 0.500 & 3.375 & 2.825 & -0.025 \\ 1.325 & 0.675 & 2.325 & 1.900 \end{bmatrix}$$

Q4 | Second Convolution Layer

$$P = \begin{bmatrix} 3.1100 & 1.625 & 2.275 & 3.400 \\ 3.900 & 4.025 & 2.100 & 2.075 \\ 0.500 & 3.375 & 2.825 & -0.025 \\ 1.325 & 0.675 & 2.325 & 1.900 \end{bmatrix}$$

Branch A (filter f_2)

$$f_2 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

* Valid 2x2 cross-correlation on P (stride=1) give 3x3 output

Top left: $\begin{bmatrix} 3.1100 & 1.625 \\ 3.900 & 4.025 \end{bmatrix} \odot \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \Rightarrow 3.1100 - 1.625 + 0 + 4.025 = 5.500$

First row, second column: $\begin{bmatrix} 1.625 & 2.275 \\ 4.025 & 2.100 \end{bmatrix} \odot \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \Rightarrow 1.625 - 2.275 + 0 + 2.100 = 1.450$

Result :

$$Y_A = \begin{bmatrix} 5.500 & 1.450 & 0.950 \\ 3.250 & 4.750 & 0.000 \\ -2.200 & 2.875 & 4.750 \end{bmatrix}$$

Branch B (filter f_3 , same padding, stride=2)

$$f_3 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

* Some padding with stride 2 on 4×4 input yields 2×2 output.

Two explicit positions:

$$\begin{aligned} \text{Top left output : } & (0.3100) + (1.1.625) + (0.2.275) \\ & + (1.3.900) + (-2.4.025) + (1.2.100) + (0.0.500) + (1.3.375) \\ & + (0.2.825) = 2.950 \end{aligned}$$

Top right output element:

$$\begin{aligned} & (0.2.275) + (1.3.400) + (0.0) + (1.2.100) + (-2.2.075) + \\ & (1.0) + (0.2.825) + (1.(-0.025)) + (0.0) = 1.325 \end{aligned}$$

Result : 2×2 matrix

$$Y_B = \begin{bmatrix} 2.950 & 1.325 \\ 5.675 & -1.500 \end{bmatrix}$$

Q5 Pooling per Branch

Branch A: min Pooling (2×2 , stride = 1 on X_A)

Output size: $(3-2)+1 = 2$ hence 2×2

Top-left block: $\min \{ 5.500, 1.450; 3.250, 4.750 \} = 1.450$

$$P_A = \begin{bmatrix} 1.450 & 0.000 \\ -2.200 & 0.000 \end{bmatrix}$$

Branch B: Average Pooling (2×2 , stride = 1 on X_B)

With 2×2 input, stride 1 yields a single value

$$P_B = \frac{2.950 + 1.325 + 5.675 + (-1.500)}{4} = 2.1125$$

$$P_B = [2.1125]$$

Q6 | Batch Normalization

- Use per-feature-map BN with $\gamma=1$, $\beta=0$, $\epsilon=10^{-2}$

$$\hat{x} = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}}, \quad y = \gamma x + \beta$$

BN on Branch A pooled map P_A

Entries = $\{1.450, 0.000, -2.200, 0.000\}$.

$$\mu_A = \frac{1.450 + 0 - 2.200 + 0}{4} = -0.1875$$

$$\sigma_A^2 = \frac{(1.450 + 0.1875)^2 + (0 + 0.1875)^2 + (-2.200 + 1.875)^2 + (0 + 0)^2}{4}$$

Denominator: $\sqrt{\sigma_A^2 + \epsilon} = \sqrt{1.7005 + 0.01} \approx 1.308$

Top-left element normalization:

$$\hat{x}_{11} = \frac{1.450 - (-0.1875)}{1.308} \approx 1.252$$

Result

$$Y_A^{BN} = \begin{bmatrix} 1.252 & 0.143 \\ -1.539 & 0.143 \end{bmatrix}$$

Q6 BN on Branch B pooled map P_B

- single value map $\{2.1125\}$

$$\mu_B = 2.1125, \sigma_B^2 = 0 \Rightarrow \hat{x} = \frac{2.1125 - 2.1125}{\sqrt{0 + \epsilon}} = 0$$

$$Y_B^{BN} = [0]$$

(with $\epsilon > 0$ the normalized output is exactly zero in 1x1 case)

Q7 tanh Activation :

Apply tanh elementwise to each batch-normalized feature map.

$$\tanh(1.252) \approx 0.849$$

$$\tanh(0.143) \approx 0.142$$

$$\tanh(-1.539) \approx -0.912$$

$$\tanh(0.143) \approx 0.142$$

Result :

$$T_A = \tanh(Y_A^{BN}) = \begin{bmatrix} 0.849 & 0.142 \\ -0.912 & 0.142 \end{bmatrix}$$

$$T_B = \tanh(Y_B^{BN}) = [0.000]$$

Q8 | Flatten and fully Connected Layer

- flatten both maps and concatenate
(row-major per map, then Branch A followed by Branch B)

$$X = [0.849, 0.142, -0.912, 0.142, 0.000]^T \in \mathbb{R}^5$$

Given

$$w_1 = [0.5, -0.4, 0.3, 0.2, -0.1], b_1 = 0.1$$

$$w_2 = [-0.2, 0.6, -0.5, 0.4, 0.2], b_2 = -0.2$$

Compute logits $z_i = w_i x + b_i$

$$\ln z_1: 0.5 \cdot 0.849 = 0.4245, 0.3 \cdot (-0.912) = -0.2736$$

$$\ln z_2: 0.6 \cdot 0.142 = 0.0852, (-0.5) \cdot (-0.912) = 0.456$$

$$\text{Rounded: } z_1 \approx 0.222 \quad z_2 \approx 0.228$$

Q9 | Softmax and Error Metrics

Softmax :

$$P_i = \frac{e^{z_i}}{e^{z_1} + e^{z_2}}, i \in \{1, 2\}$$

Numerically:

$$P_1 \approx 0.499 \quad P_2 \approx 0.501$$

a) Predicted Class : Since $P_2 > P_1$, the model predicts Class 1.

b) Losses (ground - truth label is Class 0) :
Cross - Entropy (for class 0)

$$L_{CE} = -\log p_1 \approx 0.696$$

Mean squared Error :

$$L_{MSE} = \frac{1}{2} \left[(1 - P_1)^2 + (0 - P_2)^2 \right] \approx 0.251$$

c) Brief Comment : Cross - Entropy is generally preferred to classification because it directly optimizes the log-likelihood of the correct class and yields stronger, well calibrated gradients for probability outputs, whereas MSE can lead to slower convergence and poorer probabilistic calibration for classification.